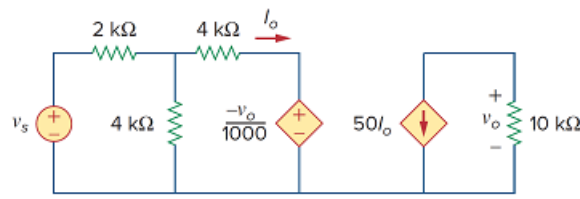


Problem

Determine the gain v_o/v_s of the transistor amplifier circuit in Fig. 3.124.

Figure 3.124:



Step-by-step solution

Step 1 of 6

Refer to the circuit in Figure 3.124 in the textbook.

Identify the loops and voltages in the circuit diagram as shown in Figure 1.

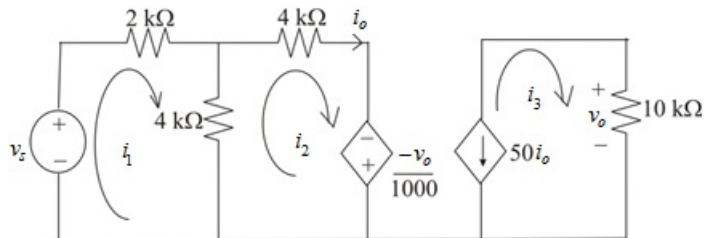


Figure 1

Step 2 of 6

Consider the expression for the voltage v_o .

$$v_o = (-50i_o)(10 \text{ k})$$

$$v_o = (-500 \text{ k})i_o$$

$$i_o = \frac{-v_o}{500 \text{ k}}$$

Therefore, the expression for the current i_o is, $i_o = \frac{-v_o}{500 \text{ k}}$.

Step 3 of 6

Apply Kirchhoff's Voltage Law in Loop-2.

$$(i_2 - i_1)(4 \text{ k}) + i_2(4 \text{ k}) - \frac{v_o}{1000} = 0$$

$$i_2(4 \text{ k} + 4 \text{ k}) - i_1(4 \text{ k}) - \frac{v_o}{1000} = 0$$

$$i_2(8 \text{ k}) - i_1(4 \text{ k}) - \frac{v_o}{1000} = 0$$

Substitute i_o for i_2 .

$$i_o(8 \text{ k}) - i_1(4 \text{ k}) - \frac{v_o}{1000} = 0$$

Step 4 of 6

Substitute $\frac{-v_o}{500 \text{ k}}$ for i_o .

$$-\left(\frac{v_o}{500 \text{ k}}\right)(8 \text{ k}) - i_1(4 \text{ k}) - \frac{v_o}{1000} = 0$$

$$-\frac{8v_o}{500} - i_1(4 \text{ k}) - \frac{v_o}{1000} = 0$$

$$\frac{-16v_o - i_1(4000 \text{ k}) - v_o}{1000} = 0$$

$$-17v_o - i_1(4000 \text{ k}) = 0$$

$$i_1(4000 \text{ k}) = -17v_o$$

$$i_1 = \frac{-17v_o}{4000 \text{ k}}$$

Therefore, the expression for the current i_1 is, $i_1 = \frac{-17v_o}{4000 \text{ k}}$.

Step 5 of 6

Apply Kirchhoff's Voltage Law in Loop-1.

$$(i_1 - i_2)(4 \text{ k}) + i_1(2 \text{ k}) - v_s = 0$$

$$i_1(4 \text{ k} + 2 \text{ k}) - i_2(4 \text{ k}) - v_s = 0$$

$$i_1(6 \text{ k}) - i_2(4 \text{ k}) - v_s = 0$$

Substitute $\frac{-17v_o}{4000 \text{ k}}$ for i_1 and i_o for i_2 .

$$\left(\frac{-17v_o}{4000 \text{ k}}\right)(6 \text{ k}) - i_o(4 \text{ k}) - v_s = 0$$

Step 6 of 6

Substitute $\frac{-v_o}{500 \text{ k}}$ for i_o .

$$\left(\frac{-17v_o}{4000 \text{ k}}\right)(6 \text{ k}) - \left(-\frac{v_o}{500 \text{ k}}\right)(4 \text{ k}) - v_s = 0$$

$$\frac{-51v_o}{2000} + \frac{4v_o}{500} - v_s = 0$$

$$\frac{-51v_o + 16v_o - 2000v_s}{2000} = 0$$

$$-51v_o + 16v_o - 2000v_s = 0$$

$$-35v_o = 2000v_s$$

$$\frac{v_o}{v_s} = -\frac{2000}{35}$$

$$= -57.14$$

Therefore, the value of the voltage gain $\frac{v_o}{v_s}$ is -57.14.